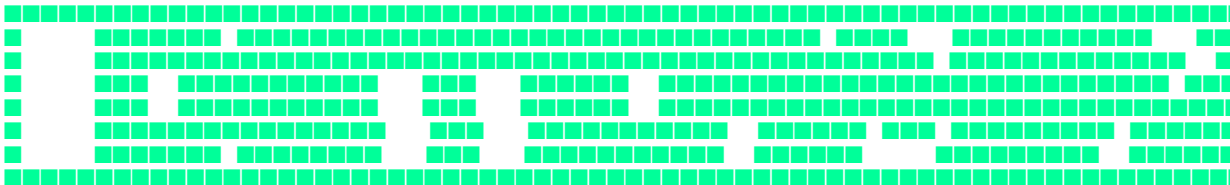


DSTERMINAL

SECURITY OPERATIONS
CENTER

Version 4.0.0.113

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■■ FOR AUTHORIZED SECURITY PERSONNEL ONLY ■■

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1. INTRODUCTION TO DSTERMINAL

DSTerminal is a comprehensive Security Operations Center (SOC) platform designed for defensive security professionals. It provides a unified terminal interface for threat detection, incident response, forensics, and system hardening operations.

Key Features:

- Real-time network monitoring with threat detection and geolocation
- AES-256 encryption for file protection and secure key management
- Memory forensics and volatile data analysis
- System hardening and vulnerability assessment
- Steganography detection in images and files
- Integration with VirusTotal API for hash lookups
- Metasploit Framework integration for penetration testing
- Automated PDF report generation for compliance
- SSL/TLS certificate analysis and security auditing
- SQL injection scanning with sqlmap integration

COMPONENT	STATUS	VERSION

2. SYSTEM REQUIREMENTS

Hardware Requirements:

Component	Minimum	Recommended

Software Requirements:

[illegible]

3. INSTALLATION GUIDE

3.1 Windows Installation

```
# DSTerminal Windows Installer

1. Download DSTerminal_Installer_v4.0.0.113.exe from GitHub
2. Run the installer (no administrator rights required)
3. Follow the setup wizard to choose installation options
4. Select components to install (Core, Documentation, Tools)
5. Choose update channel (Stable/Beta/Nightly)
6. Launch DSTerminal from Start Menu or Desktop shortcut

# Installation path
C:\Users\[USERNAME]\AppData\Roaming\DSTerminal\
C:\Users\[USERNAME]\AppData\Roaming\DSTerminal_Workspace\
```

3.2 Linux Installation

```
# Clone the DSTerminal repository
git clone https://github.com/Stark-Expo-Tech-Exchange/DSTerminal_releases_latest.git
cd DSTerminal_releases_latest

# Install Python dependencies
pip install -r requirements.txt

# Optional: Install additional tools
sudo apt update
sudo apt install nmap sqlmap ffmpeg

# Run DSTerminal
python dsterminal.py

# Create desktop shortcut (optional)
chmod +x dsterminal.desktop
cp dsterminal.desktop ~/.local/share/applications/
```

3.3 macOS Installation

```
# Install using Homebrew
brew install dsterminal

# Or from source
git clone https://github.com/Stark-Expo-Tech-Exchange/DSTerminal_releases_latest.git
cd DSTerminal_releases_latest
pip3 install -r requirements.txt
python3 dsterminal.py
```

4. GETTING STARTED

4.1 First Launch

```
[22:41:49] WILSON-SON [PROD] NORMAL [SOC-5BD39]
■ OP-189EA9 @ soc-terminal : ~$ help
```

```

██████████DSTERMIAL HELP MENU██████████

```

4.2 Workspace Initialization

```
# Navigate to DSTerminal workspace
cd ~/DSTerminal_Workspace

# Create standard directories
mkdir operators
mkdir scans
mkdir reports
mkdir exploits
mkdir sandbox
mkdir quarantine

# Verify workspace structure
ls -la
```

5. CORE COMMANDS

5.1 File Operations

ls [path]	List directory contents
cd <directory>	Change working directory
pwd	Print current working directory
mkdir <name>	Create new directory
touch <file>	Create empty file
cat <file>	Display file contents
echo <text>	Display text
echo <text> > <file>	Write text to file
echo <text> >> <file>	Append text to file

5.2 System Commands

sysinfo	Display detailed system information
killproc <PID>	Terminate a process by ID
update	Check for DSTerminal updates
clear	Clear the terminal screen
exit	Exit DSTerminal
help	Show comprehensive help menu
status	Display DSTerminal system status

6. SECURITY MODULES

6.1 System Security Scan

```
system scan -All [--quick] [--deep] [--output <file>]
```

Performs comprehensive DSTerminal security scan including:

- Memory analysis and process inspection
- File integrity verification
- Network connection monitoring
- Registry analysis (Windows systems)
- Scheduled tasks and services review
- Malware persistence mechanism detection

6.2 Exploit Detection

```
exploitcheck
```

Checks for critical vulnerabilities:

- CVE-2021-44228 - Log4j Remote Code Execution
- CVE-2017-0144 - EternalBlue SMB Exploit
- CVE-2021-3156 - Sudo Buffer Overflow
- Local privilege escalation vectors
- Kernel exploit detection

6.3 Integrity Verification

```
check integrity
```

Verifies critical system files:

- Windows: kernel32.dll, cmd.exe, powershell.exe
- Linux: /bin/bash, /usr/bin/sudo, /bin/ls
- macOS: /bin/bash, /usr/bin/sudo
- Checks file sizes, modification times, and hashes

7. NETWORK OPERATIONS

7.1 Real-time Network Monitor

```
net -n mon [--duration <sec>] [--export <file>] [--geoip]
```

DSTerminal Network Monitoring Features:

- Live connection tracking with ESTABLISHED/LISTEN states
- Bandwidth usage monitoring (upload/download)
- Geolocation of remote IP addresses
- Threat scoring system (LOW/MEDIUM/HIGH)
- Automated PDF forensic report generation
- Intrusion detection alerts for port scans
- ISP and country information for connections

Threat Score	Risk Level	Indicator
	LOW	
	MEDIUM	
	HIGH	

7.2 Port Scanning

```
portsweep <target> [--ports range]
```

DSTerminal port scanner scans:

- Common ports: 21,22,23,25,53,80,110,135,139,143
- Additional ports: 443,445,3389,8080,8443
- Custom port ranges (e.g., --ports 1-1024)
- Service identification when possible

7.3 Traceroute

```
traceroute <target>
```

Network path analysis:

- Traces route to target destination
- Shows intermediate hops
- Response time measurements

8. CRYPTOGRAPHY

8.1 DSTerminal Encryption System

```
# Initialize encryption (first time only)
encrypt-setup

# Encrypt a file
encrypt <filename>

# Decrypt a file
decrypt <filename.enc> [key]

# List encrypted files
crypto-list

# Show encryption information
crypto-info <file.enc>

# Verify encryption system
crypto-verify

# Backup encryption key
crypto-backup

# Test encryption system
encrypt-test
```

- • AES-256 encryption using Fernet implementation
- • Automatic key generation and secure storage (~/.dsterminal_key)
- • Secure file deletion with multiple overwrite passes
- • File integrity verification with SHA-256 hashing
- • QR code key backup for offline storage
- • Encrypted file inventory management
- • Key ID tracking for multiple keys

9. FORENSICS

9.1 Memory Forensics

```
memdump [--output <file>]
```

DSTerminal Memory Forensics captures:

- Running processes and their memory maps
- Active network connections
- Open file handles
- Loaded kernel modules
- Windows registry hives
- Process environment variables

9.2 Steganography Detection

```
stegcheck <image_path>
```

DSTerminal Steganalysis includes:

- Least Significant Bit (LSB) analysis
- Entropy calculation and evaluation
- Known steganography tool signatures
- EXIF metadata examination
- File structure anomaly detection

9.3 Ransomware Detection

```
ransomwatch
```

DSTerminal Ransomware Scanner detects:

- Suspicious file extensions (.encrypted, .locked, .crypt)
- Mass file encryption patterns
- Ransom note creation
- Bitcoin wallet address generation
- Process behavior anomalies

10. SYSTEM HARDENING

10.1 DSTerminal Hardening

```
hardener -t sys [--dry-run]
```

DSTerminal Hardening applies:

- Windows: Disables SMB1 protocol
- Windows: Configures firewall rules
- Linux: Enables UFW firewall
- Linux: Secures SSH configuration
- Disables unnecessary services
- Applies CIS benchmark recommendations
- Configures system auditing

10.2 MAC Address Spoofing

```
macspoof [interface]
```

DSTerminal MAC Changer:

- Generates random valid MAC addresses
- Auto-detects active network interface
- Supports Linux, Windows, and macOS
- Verifies MAC change success
- Progress animation during spoofing

10.3 Log Management

```
clearlogs
```

DSTerminal Log Cleaner:

- Windows: Application, System, Security event logs
- Linux: /var/log/* files
- Systemd journal (journalctl)
- Shell history files
- Application-specific logs

11. INCIDENT RESPONSE

11.1 Financial Attack Simulator

```
transfertrace  
extract -money
```

```
DSTerminal Financial Simulator:  
• SWIFT network transaction simulation  
• Cryptocurrency exchange monitoring  
• Payment processor analysis  
• Money laundering pattern detection  
• Transaction visualization  
• Suspicious activity reporting
```

11.2 Emergency Procedures

```
shutdown
```

```
DSTerminal Emergency Shutdown:  
• Requires explicit confirmation ('YES')  
• 15-second countdown with visual indicator  
• Forces system shutdown  
• Supports Windows and Linux
```

12. REPORTING

[illegible]

13. TROUBLESHOOTING

[illegible]

14. APPENDIX

14.1 DSTerminal Directory Structure

DSTerminal Installation Directory:

- dsterminal.exe (Main executable)
- config/ (Configuration files)
- docs/ (Documentation)
- tools/ (Helper utilities)
- templates/ (Report templates)
- logs/ (Application logs)
- updates/ (Downloaded updates)

DSTerminal Workspace:

- operators/ (Session logs)
- scans/ (Scan results)
- reports/ (Generated reports)
- exploits/ (Safe exploit storage)
- sandbox/ (Testing environment)
- quarantine/ (Isolated threats)

14.2 API Keys Configuration

```
# VirusTotal API Key (optional)
DSTerminal supports VirusTotal integration:
1. Get API key from virustotal.com
2. Edit CONFIG in dsterminal.py
3. Use 'vt-scan' command for hash lookups

# GitHub Token (for updates)
Configure in CONFIG['UPDATE_URL']
```

14.3 Update Channels

DSTerminal Update Channels:

- Stable Channel: Production-ready releases
- Beta Channel: Early access to new features
- Nightly Channel: Daily development builds

Select during installation or in config/update_channel.conf

14.4 Support and Community

GitHub Repository	https://github.com/Stark-Expo-Tech-Exchange/DSTerminal_releases_latest
Documentation	https://starkexpotechexchange-mw.com/docs
Issue Tracker	https://github.com/Stark-Expo-Tech-Exchange/DSTerminal_releases_latest/issues
Email Support	support@starkexpotechexchange-mw.com